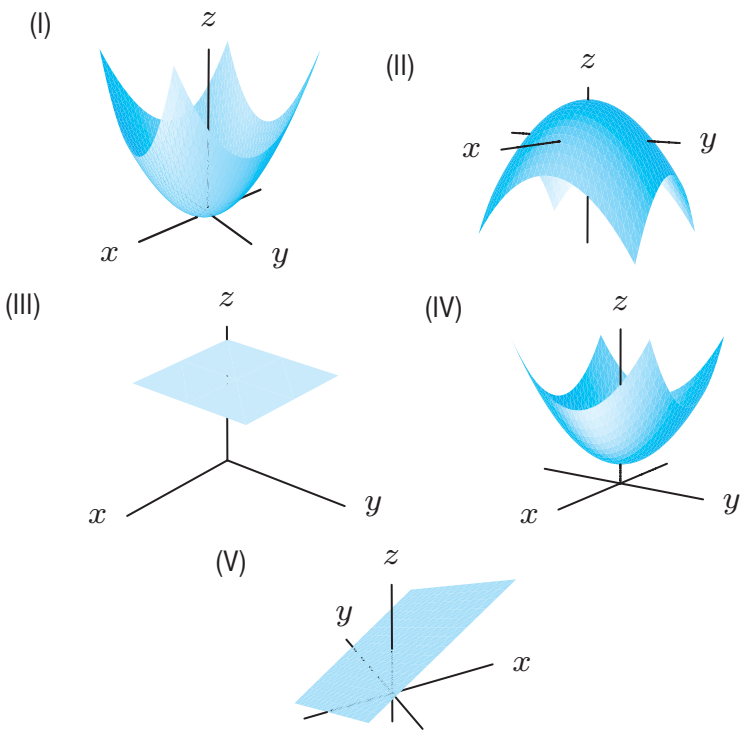
Suggested problems in §12.2: 1-13 odd, 19

Suggested problems in §12.3: 5-11 odd, 17

Problem 1: Match the graphs with the functions.



(a) $z = 2 + x^2 + y^2$ IV

(b) $z = 2 - x^2 - y^2$ II

(c) $z = 2(x^2 + y^2)$ I

(d) $z = 2 + 2x - y$ V

(e) $z = 2$ III

Problem 2: Consider the function $f(x, y) = y^3 + xy$. Draw cross sections of f with:

(a) x fixed at $x = -1$, $x = 0$, and $x = 1$

See class notes.

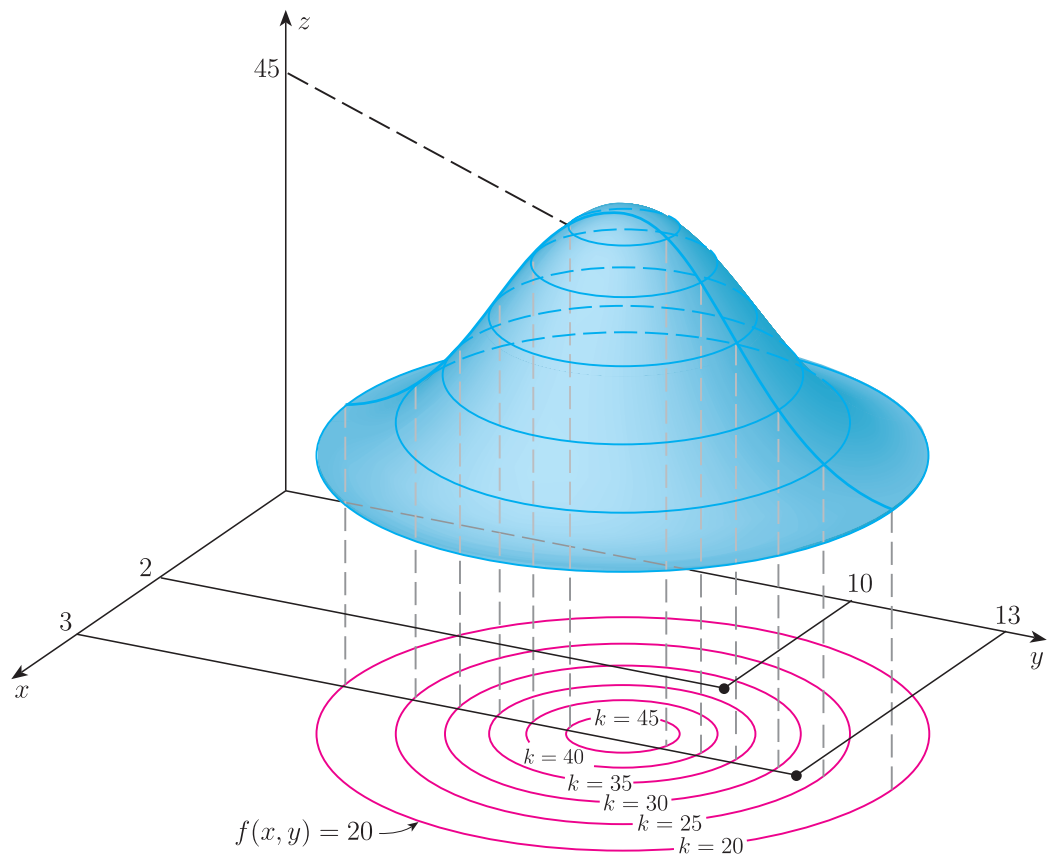
(b) y fixed at $y = -1$, $y = 0$, and $y = 1$

See class notes.

Problem 3: Plot the surface in $\mathbb{R}^3$ defined by the equation $y^2 + z^2 = 1$.

See class notes for the drawing. In words, this is an infinitely long cylinder of radius 1 that wraps around the x-axis.

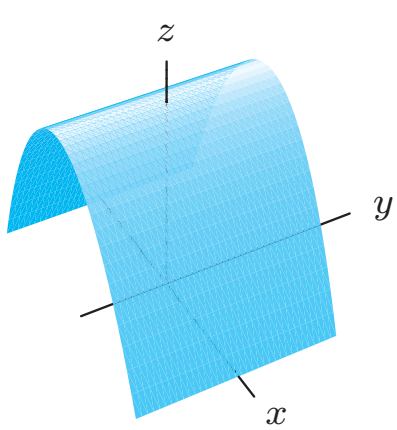
Problem 4: The picture below shows a contour map of a function $z = f(x, y)$. Use the contour map to estimate the given values of $f(x, y)$.



(a) $f(2, 10) \approx$ 32.5

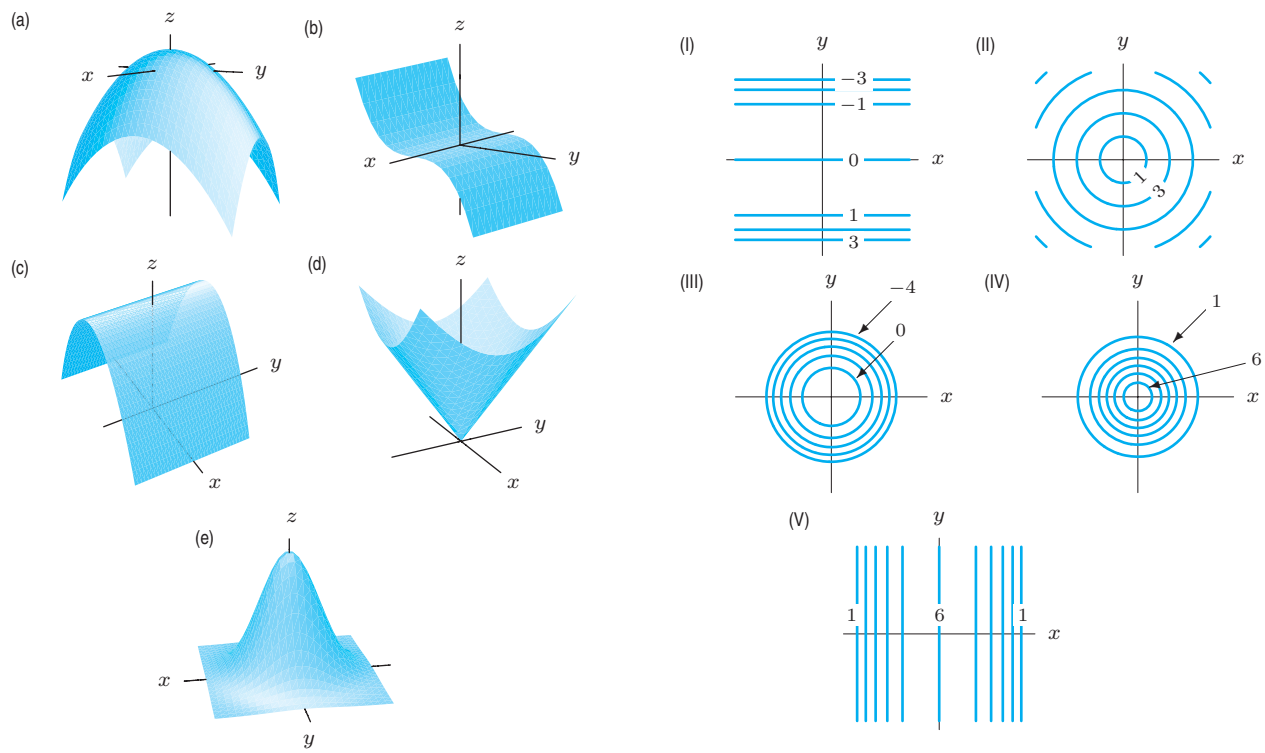
(b) $f(3, 13) \approx$ 27.5

Problem 5: Sketch a possible contour diagram for the following surface.



See class notes

Problem 6: Match the surfaces to the contour diagrams.



(a) and III, (b) and I, (c) and V, (d) and II, (e) and IV.

Problem 7: Sketch a contour diagram for the graph of $f(x, y) = xy$.

See class notes.